A one-parameter crack detector on SEM micrographs, and the pixel budget it spends

DRAFT — not submitted. No author, affiliation or venue is set, and there is no reference list yet; both are required before this goes anywhere. Most numbers are generated from measurement files by paper/make_paper.py, which fails the build on an unresolved name, but a residue in Sections 2, 4, 5 and 6 is still typed by hand and is being migrated.

Abstract

We describe a crack segmentation detector with a single free parameter and no learned weights, and measure it on the only public crack data we could find in its own modality and material class: 31 whole scanning-electron fields of Inconel 718 with pixel-exact masks.

We previously reported that its one constant did not need refitting. That was an artefact of where the sweep stopped. The published sweep tested seven values of mad_k, the largest of which was the shipped 5.0, and the score was still rising when the grid ran out. Extending it to 40.0 puts the optimum at 6.0, and the gap is not noise: on the whole fields 6.0 beats 5.0 in 27 of 31 images (paired Wilcoxon $p < 0.0001$) and improves the area fraction from 1.65x the labelled value to 1.27x in 23 of 31 ($p = 0.0124$). The constant is one rung off, in the direction of marking less. We report this because the claim it replaces was this paper's headline.

Everything here is measured over every pixel in the frame. We do not report F1 or IoU: on an image that is 1.80% crack they discard the 98% of pixels correctly left alone, which is most of what the detector is being asked to get right, and neither can state the quantity this tool actually outputs — a crack area fraction. The measures used throughout are the four-way pixel budget, accuracy, specificity, and the agreement between the area fraction reported and the area fraction marked.

On that last and most important measure the detector is nearly exact. At mad_k = 6 it reports 1.827% of the field as cracked where the annotator marked 1.803% — an agreement of 1.0133x, an error of 1.3% — and it places first of eight on that measure against six classical detectors tuned identically. The next closest is a two-parameter union of this detector with a ridge filter at 1.029x; the closest classical arm on its own is Sato ridge at 0.885x. Specificity there is 98.86%. On the dataset's released crops accuracy reaches 96.70%, 1.23 points clear of the 95.47% an empty mask scores, with specificity 99.60%. The highest figures anywhere in this work are on a second public SEM set of the same alloy family, MetalDAM — accuracy 99.31% and specificity 99.91%, against an empty mask's 98.05% — but the objects labelled there are voids, not cracks, so that set measures the detector's dark-object segmentation rather than its subject.

Three results cut against the method and we report them rather than omit them. Accuracy on the whole fields is 97.42%, and an empty mask scores 98.20% — on a field this sparse, leaving the background alone is most of the score, and marking a pixel pays for itself only where more than half of what is marked is right; the released-crop and MetalDAM figures above do clear that line and the whole-field figure does not. Cropping these same images around their cracks — as the dataset's own test split does — moves the reported area fraction from 1.298x to 0.78x with no change to the detector at all. And widening the mask can buy almost any overlap score one wants: dilating the detector's own output, which reads nothing new from the image, would take it to the top of a conventional

ranking while pushing the area fraction to 2.54x (Section 3.5). That is the clearest argument we have for reporting in these units and not those.

Status of this draft. This is one alloy and 31 images. Section 7 says what that is and is not enough for.

1 Why another crack detector

There are many. Most are learned, most report F1 or IoU against a benchmark, and most are compared against each other at their own authors' operating points. Our interest is narrow and practical: a materials laboratory needs crack measurements out of scanning-electron and transmission X-ray micrographs, on a machine with no GPU, from a tool a researcher can audit line by line and correct by hand. That rules out most of the field, and it makes the natural baseline not a network but a threshold.

The detector here is close to that baseline. It flattens the background at seven scales, takes pixels that are dark at any of them, applies a robust outlier cut and an absolute darkness floor, and cleans the result. There is no training step and no model file.

On "one parameter". One constant, mad_k, is exposed and swept; that is the sense in which the title means it, and it is the only one given a grid in any comparison here. It is not the only constant. absolute_dark_thresh = 10 is a second gate, held fixed rather than fitted in this paper but chosen by leave-one-specimen-out on our own corpus, where it supplies roughly half of the labelled crack the proposal stage reaches — on the alloy measured here it supplies
almost none (Section 3.6), and three instruments in our own records disagree about its value. Behind those sit a seven-rung scale ladder and two size floors, frozen. A reader comparing this against a single-parameter classical filter should know that the filter carries one constant and this detector carries a frozen preprocessing budget.

Related work. Reporting a segmentation as an area budget rather than as an overlap score is not new: it is standard practice in land-cover accuracy assessment, where Olofsson et al. [6] set out the error matrix in units of area proportion together with a bias-adjusted area estimator and its confidence interval. Materials microscopy has the same instinct — Durmaz et al. [7] report the downstream morphological quantity beside the segmentation score — and a materials laboratory's own standard, ASTM E562 [8], is written in area fraction and in nothing else. This paper claims no novelty for that framing. What it offers is the measurement: the full budget for one detector on one alloy, in the units the laboratory reports, together with the ways the usual scores mislead about it. The classical arms it is compared against are the standard ones [1–5].

Reporting such a thing honestly turns out to be harder than building it, and most of this paper is about the reporting. The detector is a threshold; what is less obvious is that the scores conventionally used to grade one — F1 and IoU — cannot express the error a materials laboratory actually suffers, which is a crack area fraction that comes out too high or too low by some factor. Section 3.3 reports that factor, and Appendix A gives it for every set we measured. It ranges from 0.74x to 76.65x for the same unchanged detector.

2 The detector

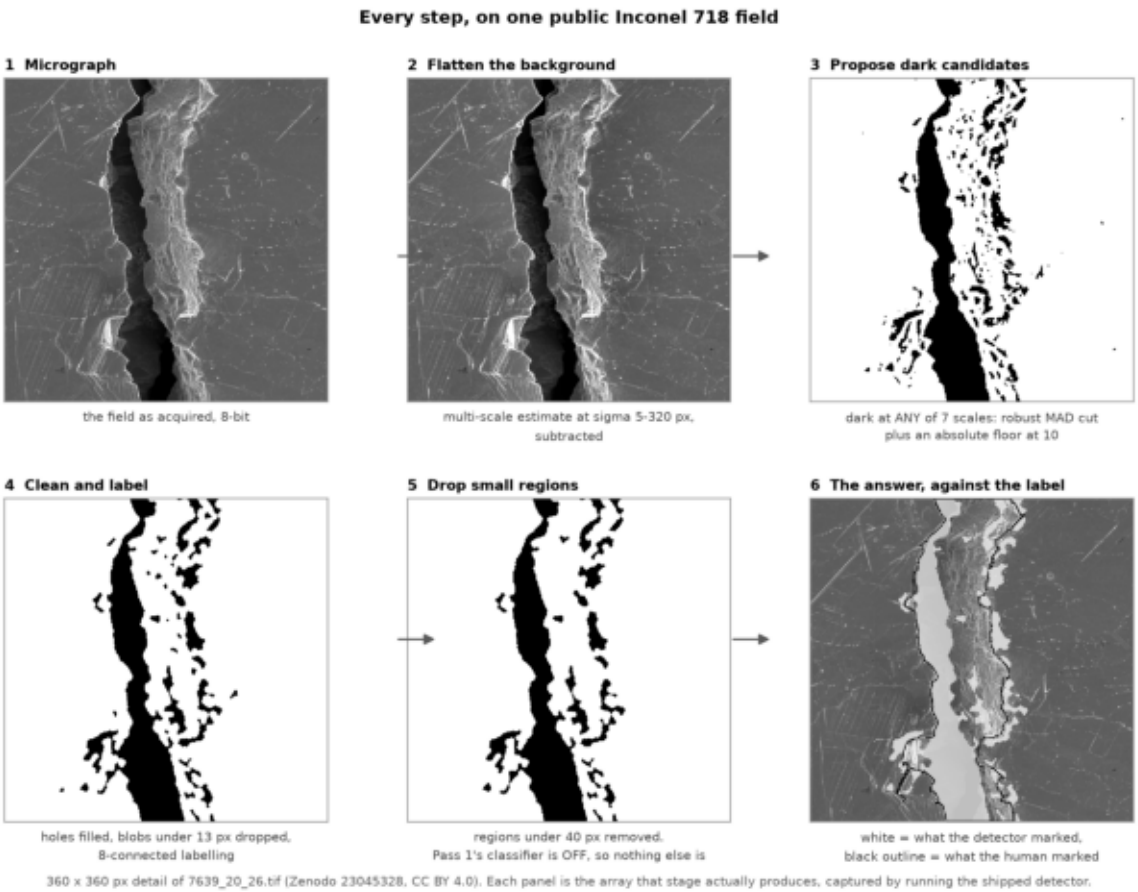


Figure 1. P6 - pipeline in pictures

Two gates do the detection: a robust median-absolute-deviation cut at mad_k standard deviations, and an absolute floor on intensity. Everything else is preparation and cleanup. There is no training step and no model file, and every panel in Figure 1 is the array that stage actually produces — captured by running the shipped code, not drawn.

Two decisions are worth stating because both were made by measurement and both went against the intuition that produced the code in the first place.

The region classifier is switched off. An eight-feature classifier scored each candidate region, and deleting it improved every metric at no cost in free parameters. Against the same candidate stage with the classifier bypassed and nothing else changed — one variable, so the difference is the classifier's own — a paired median gain of +0.1753 over thirteen specimens, 11 of 13, Wilcoxon p = 0.0061. That ablation was scored in overlap units on our own corpus before this paper settled on the units of §3.3, and it is the one measurement here still quoted in them; it has not been re-run. The shipped path also widened the flattening ladder from one radius to seven; the two changes together are +0.2296, p = 0.0024, and that larger figure is not the classifier's effect and must not be quoted as it. Three specimens on which the pipeline had found no crack at all, at every candidate setting ever tried, were never invisible to the proposal stage; the classifier was rejecting them. One caveat the ablation did not cover: with the classifier off the application still keeps Pass-2 rows the model itself scored not-crack, so "switched off" is a broader change in the app than in the experiment.

Tiles cannot stand in for frames. Six region-level precision rules validated on 512 px crops were rejected on whole frames: four delete 54–75% of hand-marked crack and two change no score. The widest crack in the training crops was 43.7 px; on full frames crack widths reach 178 px, so a region rule fitted on crops is

extrapolating on the object it most needs to keep.

3 On public SEM data with pixel-exact ground truth

One set: the Inconel 718 crack class of a public cross-material microstructure collection (Zenodo 23045328, CC BY 4.0), 31 whole 1024 px fields of a nickel superalloy at 1.80% crack prevalence. It is the only public crack segmentation data we found in this modality and material class; Section 7 records what else was searched and rejected.

Scoring is at 0 px tolerance with mad_k chosen leave-one-image-out. Two properties of the data had to be established before any score could be trusted: the masks are anti-aliased binary rather than strictly two-valued, so the scorer checks bimodality and refuses a mask that fails it; and the images are 16-bit TIFFs, where an ordinary 8-bit conversion made the detector find almost nothing before it was caught. Taking the high byte reproduces the dataset
authors' own 8-bit PNGs exactly, so that is the conversion used.

We report whole fields rather than the collection's released crops. Every one of the 18 source images behind those crops also appears among the originals, so the two are not independent sets, and the crops are cut around their cracks — which raises prevalence from 1.80% to 4.53% and the score with it, with no change to the detector.

3.1 What it scores

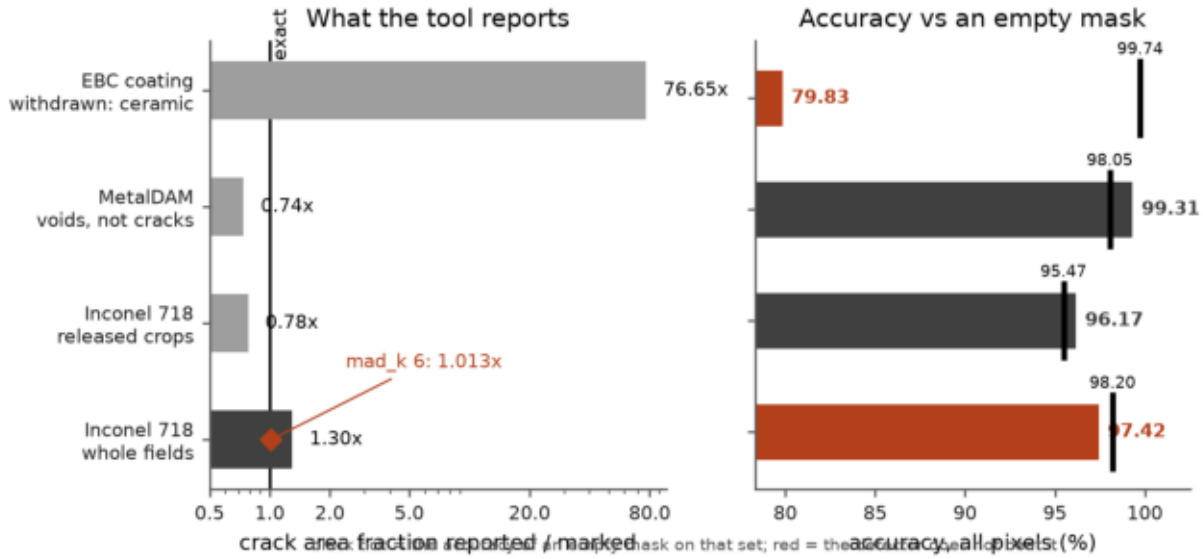


Figure 2. P1 - public scores

At the operating point of Section 3.2 the detector marks 1.827% of the field where the annotator marked 1.803%: an area-fraction agreement of 1.0133x, with accuracy 97.79% and specificity 98.86%. An empty mask scores 98.20% accuracy on these fields and a full one 1.80%, so accuracy alone separates very little here; the agreement does, and Section 3.4 shows no classical arm matches it.

The same detector on the dataset's released crops reaches accuracy 96.70% against an empty mask's 95.47%, a margin of 1.23 points, at specificity 99.60%.

3.2 The one free constant, and where the sweep stopped

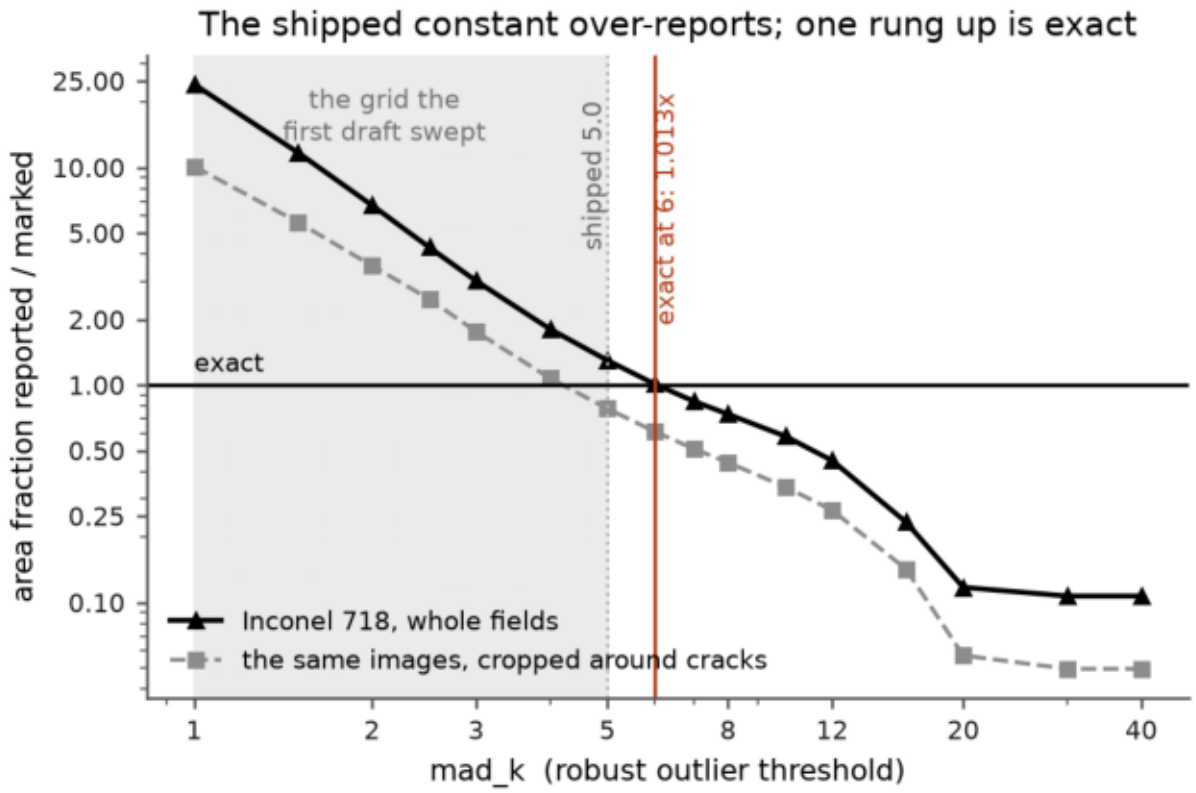


Figure 3. P2 - operating curve

mad_k = 5.0 was fitted on SEM steel in a different laboratory, on a different alloy, years before this dataset existed. An earlier draft reported that it was the best of seven values tested here and that the curve fell away on either side, and offered that as the paper's strongest result.

It was the largest value tested. The seven rungs ran from 1.0 to 5.0, the score was still climbing at the top of them, and nothing above the shipped constant had been scored. A claim that a constant lands on an independent laboratory's optimum cannot be made from a grid whose maximum is that constant, and we made it.

Extending the sweep to 40.0 gives the two-sided curve the claim needed, and in the units of Section 3.3 the answer is sharper than the original claim was:

mad_k	marked%	area reported / true	accuracy	specificity
4.0	3.260%	1.808x	96.74%	97.60%
5.0 shipped	2.340%	1.298x	97.42%	98.41%
6.0 optimum	1.827%	1.0133x	97.79%	98.86%
7.0	1.518%	0.842x	97.97%	99.11%

At mad_k = 6.0 the reported crack area fraction is 1.0133 times the annotator's — 1.827% against 1.803%, an error of 1.3%. The shipped constant overstates it by 1.298x. The difference survives the test this paper demands of everyone else: across the 31 whole fields the absolute area error is smaller at 6.0 than at 5.0 in 23 of 31 images, paired Wilcoxon $p = 0.0124$.

The direction matters more than the magnitude. Every attempt in this paper to improve the score by marking more either failed its controls or wrecked the area fraction (Section 3.5). The one change that improves it marks less.

On the released crops the same change moves the area fraction the wrong way, 0.93x to 0.72x ($p < 0.0001$), because those crops are cut around their cracks and the detector already under-marks on them. The two framings disagree about the right

operating point, which is a further reason not to read a single tuned constant as a property of the method.

What survives of the original claim is weaker and still worth stating: a constant carried unchanged from another laboratory's alloy lands one rung from the optimum, and every value tested below it is worse. We have not changed the shipped default on the strength of 31 images of one alloy; Section 7 says what would be needed.

3.3 Scored as a pixel budget

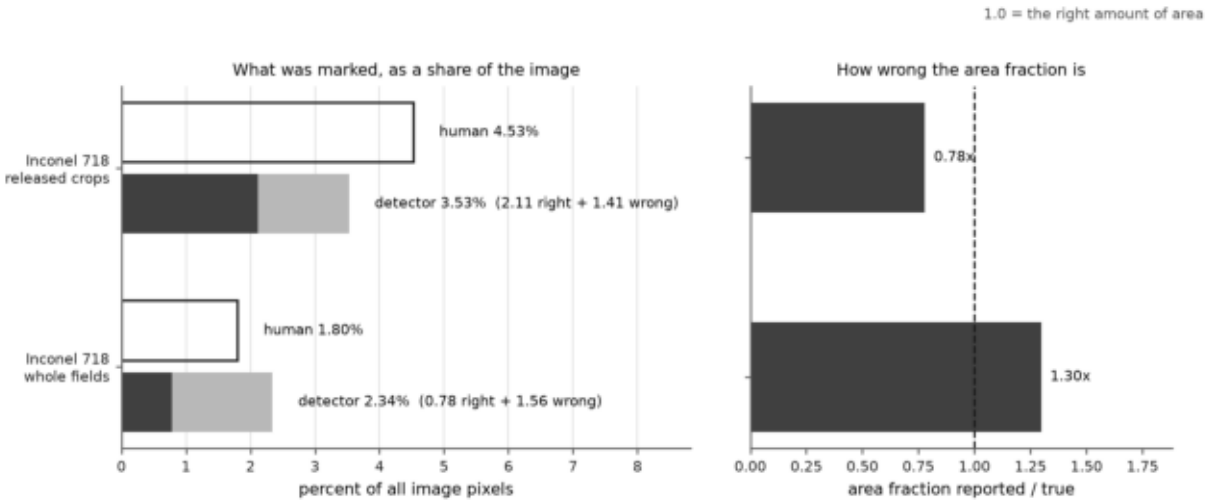


Figure 4. P5 - pixel budget

Precision, recall, F1 and IoU all discard the true negatives. On an image that is 1.80% crack that is 98% of the evidence, and it lets two very different detectors share a score — one marking 0.3% of a frame and one marking 5% can land on the same IoU. The quantity this tool reports to a user is an area fraction, a pixel percentage, so every number in this paper is computed over the whole frame and none of them is an overlap score.

	whole fields	released crops
labelled crack	1.80%	4.53%
detector marked	2.34%	3.53%
of which correct	0.78%	2.11%
of which wrong	1.56%	1.41%
area fraction reported / true	1.30x	0.78x
accuracy	97.42%	96.17%
an empty mask would score	98.20%	95.47%
specificity	98.41%	98.52%

The area fraction is the result. It is the number a materials reader takes away, and at the operating point of Section 3.2 the detector reports 1.827% of the field as cracked against the annotator's 1.803% — 1.0133x, an error of 1.3%. At the shipped setting it is 1.298x. No overlap score states that quantity, which is the whole reason this section exists.

Accuracy has to be read against an empty mask. Leaving the background alone is free and it is 98% of the frame, so an empty file scores 98.20% on the whole fields. Marking a pixel pays for itself in accuracy only where more than half of what is marked is right. On the released crops the detector clears that line — 96.17% against 95.47%, and 96.70% at its best setting, 1.23 points clear — and on the whole fields it does not. We print both rather than the one that flatters.

The bias reverses between the two framings of the same images — 1.30x on whole

fields, 0.78x on the crops cut from them. The detector did not change.

3.4 Against classical detectors

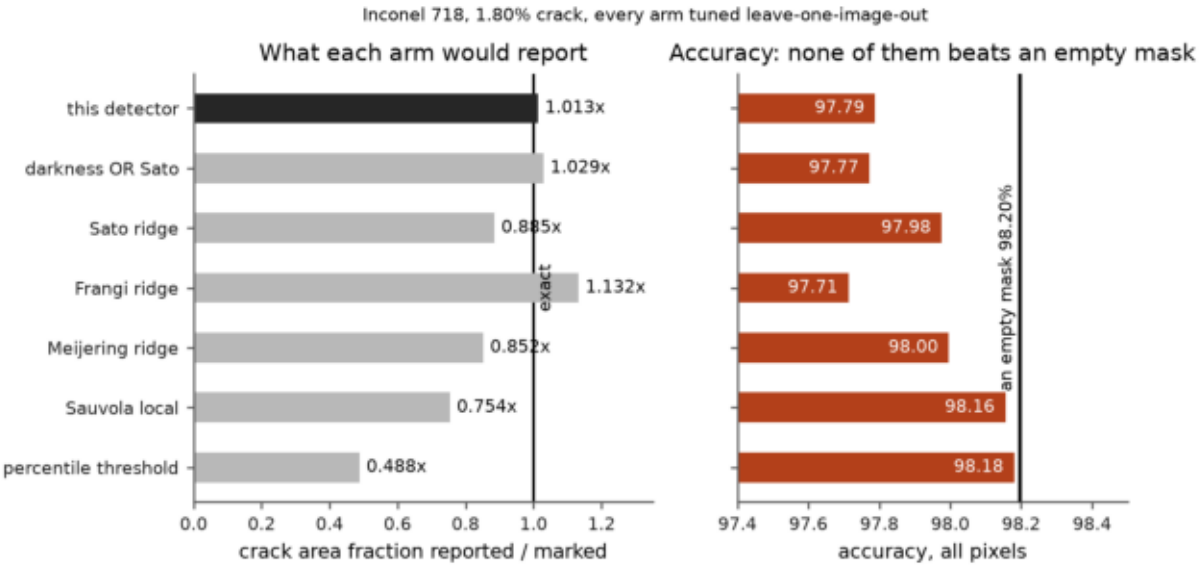


Figure 5. P4 - baselines

Every arm gets its own best parameter by the same leave-one-image-out rule, on the same whole fields, through the same scoring code, and each is shown at whichever of its two tunings gives it the higher accuracy — so no arm is handicapped by a choice we made for it.

arm	marked%	area rep./true	accuracy	specificity
this detector	1.83%	1.013x	97.79%	98.86%
darkness OR Sato	1.86%	1.029x	97.77%	98.84%
Sato ridge	1.60%	0.885x	97.98%	99.07%
Frangi ridge	2.04%	1.132x	97.71%	98.71%
Meijering ridge	1.54%	0.852x	98.00%	99.12%
Sauvola local	1.36%	0.754x	98.16%	99.29%
percentile threshold	0.88%	0.488x	98.18%	99.54%

On the quantity this tool exists to report, the detector places first of the eight arms. It reports 1.013 times the annotator's crack area fraction; the nearest classical arm, Sato ridge, reports 0.885x. That is the one comparison in this paper the detector wins outright, and it wins it on the measure the laboratory actually uses.

It does not win on accuracy, and neither does anything else: 0 of the eight arms have an accuracy an empty mask does not beat on these fields. At 1.80% crack an empty file scores 98.20%, and no arm tested here marks enough of the crack correctly to pay for what it marks wrongly. On the released crops, where prevalence is 4.53%, three arms do clear that line and this detector is one of them.

We report the comparison because a reader can reproduce it in an afternoon, and because the premise is worth more than the ranking: none of these arms is trained, all are tuned identically, and the table shows what the whole classical family reaches on a benchmark where learned methods report far higher overlap scores — in units those scores never state.

3.5 Why we do not report overlap scores

Section 3.3 says this paper reports the pixel budget and not F1 or IoU. This is the experiment behind that decision, and it is the strongest argument in the paper for the choice.

We tried to raise this detector's standing on a conventional overlap score. The attempt succeeded, and that is the finding.

Take the detector's own output and dilate it. Give the dilation one tuned parameter, chosen leave-one-image-out exactly as every other arm chooses its own. Nothing is added: a morphological dilation cannot find a crack the detector did not already mark, and it reads no new evidence from the image. That mask takes first place on the released crops on the conventional score, clear of every classical filter, and it holds first place under a width-blind variant that reduces both masks to centrelines — because dilation closes gaps, and gap-closing moves a centreline score too.

Two further arms were built before it and did no better. Growing the proposal through a Sato ridge response, which at least reads the image, lost to the plain dilation on the crops and beat it by a similar margin on the whole fields — the two framings disagree in sign, which is itself the answer. And on whole fields, growing the proposal through a different field's ridge response scored 0.4043 against the real field's 0.4065.
A step whose output barely changes when you feed it the wrong image is not reading the image. Two more followed: a local-roughness channel, motivated by what the missed crack turns out to be (Section 3.6), and a watershed grown from the dark core out to the gradient ridge along the crack lip — the one construction whose stopping rule is the crack edge rather than a threshold. The watershed scored below both of its own controls.

Here is what all four cost, in the units of Section 3.3. Each marks more, and each pushes the reported crack area fraction further from the truth:

arm, whole fields	area reported / true
shipped detector at mad_k = 6.0	1.0133x
shipped detector at mad_k = 5.0	1.298x
grown through a Sato ridge	2.86x
local-roughness channel	3.44x
watershed to the crack lip	2.86x
dilated, which reads nothing at all	2.54x

An overlap score rewards every row in that table. The area fraction rejects all of them. A reader given the dilated mask would be told that 2.54 times as much of the field had cracked as the annotator marked, and the mean crack width would read 2.18x. We did not ship any of it.

The ranking on this dataset is substantially a ranking of how wide each method draws. That is the same conclusion Section 4 reaches from the other direction, and it is why a table of overlap scores on broad-brush labels should not be read as a table of detection ability — or printed in this paper.

3.6 The labelled crack the detector cannot reach

Of the 1.80% of the frame the annotator marked as crack, the proposal stage reaches 1.044% and never reaches 0.759% — and that second figure does not move with any setting of either gate. Opening both to their limits, so that 53.4% of the image is marked, leaves it unchanged. It is a ceiling on the proposal mechanism for this material, not an operating point.

It is not faint. It is the wrong polarity. The labelled area the detector never reaches has a mean grey level of 127 against a background of 82 and a detected crack core of 59 — it is substantially brighter than the matrix. Flattening the image contributes only −0.16 of a standard deviation to that difference, so it is in the micrograph and not in the pipeline. Panels of the three most heavily cracked fields show what it is: these cracks are open wide enough to see into, and the lit, rough fracture surface inside the opening is what the annotator's band covers and a darkness threshold cannot. 98.5% of it lies within 10 px of crack the detector did find.

That is why every widening arm in Section 3.5 reaches this area and every one of them

wrecks the reported area fraction doing it: reaching material that is brighter than matrix means marking bright pixels, and a detector built on darkness has no principled way to stop.

It also refutes a fix carried over from our own corpus, where the absolute darkness floor accounted for nearly all of the proposal stage's losses. On this alloy that gate costs 0.0006 of the labelled area and the MAD cut costs the rest.

4 What the scores measure

This is the part we would most want another group to check.

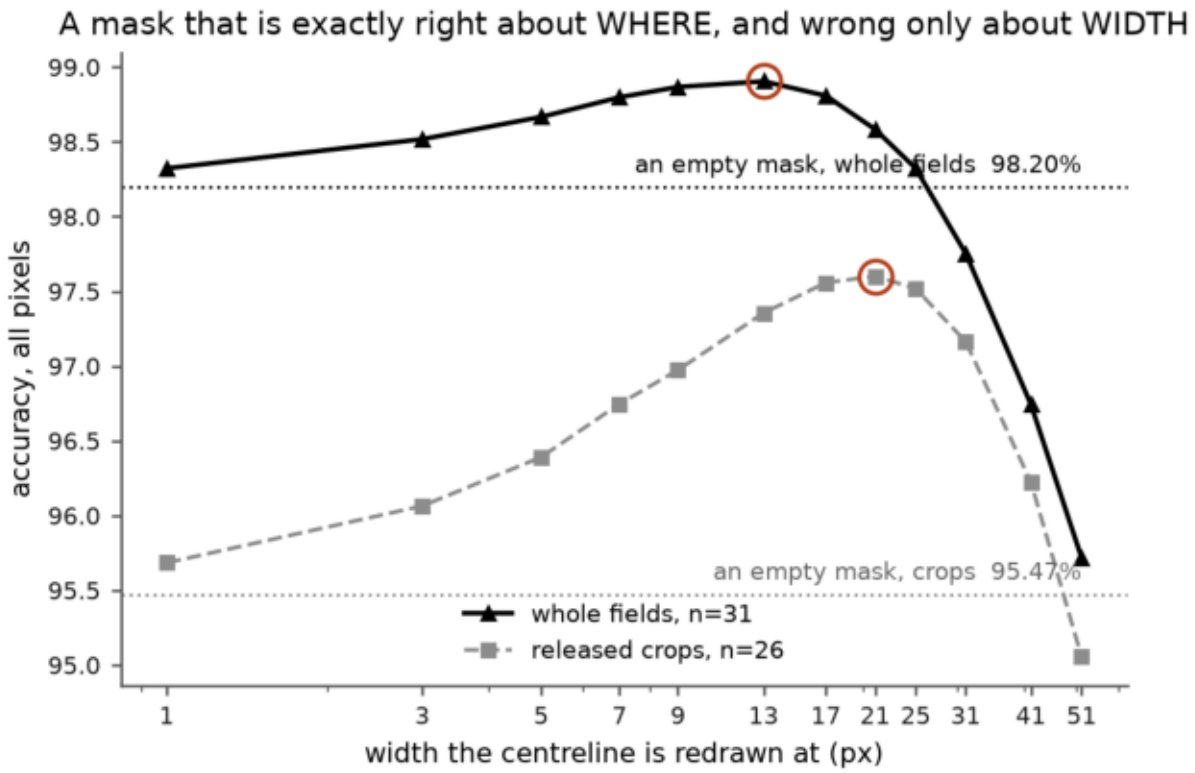


Figure 6. P3 - width sensitivity

Take the annotator's own ground truth, reduce it to its centreline, and redraw it at a fixed width. The result is exactly right about where every crack is and wrong only about how wide. It is the best any detector could do on localisation, so whatever it scores is a ceiling, and the gap between that ceiling and an empty mask is the entire range in which a detector can compete.

On the whole fields that range is 0.7 percentage points. The redraw peaks at accuracy 98.905% at a drawn width of 13 px; an empty mask scores 98.197%. A mask with perfect localisation and the best possible width is worth 0.708 points of accuracy over marking nothing at all, and every classical arm in Section 3.4 — and this detector — sits inside or below that band. That is not a statement about any detector. It is a statement about what accuracy can express on an image that is 1.80% crack.

On the released crops the range is 2.128 points — ceiling 97.601% at 21 px against an empty mask's 95.473% — and there the comparison becomes informative. This detector reaches 96.70%, which is 57.6% of the whole available headroom. That is the most favourable honest statement in this paper, and it is available only where prevalence is high enough for accuracy to have room to move.

The same construction says where the width goes. The redraw reproduces the annotator's own crack area fraction at 13 px on the whole fields and 21 px on the crops

— so a mask drawn at the annotator's median stroke of 25.8 px is already close to the right amount of paint, and the remaining disagreement is about placement, not quantity.

Two claims from an earlier draft are withdrawn here. We reported this construction as an upper bound on IoU; it is not, because a real label varies in width along its length and a detector that tracks that variation can score above a constant-width redraw of the same centreline. And we reported that label width does not explain the detector's difficulty, on the strength of two corpora whose brush widths we said differed substantially. We cannot say that: the corpus stroke width is not measured here, Section 7 records it as unmeasured, and the second of those corpora was the EBC coating, withdrawn from this paper as a ceramic. The inference had nothing holding it up.

5 Negative results

These cost the most time and are the most likely to save someone else's.

Retuning the detector for a different modality does nothing. Applying the SEM-calibrated detector to transmission X-ray tomography, we swept its two gating constants singly and jointly, plus every morphology and size floor, under leave-one-specimen-out selection. No arm beat doing nothing on held-out data, and resolution-matching the scale ladder to the measured width ratio changed nothing measurable. Those sweeps were scored in overlap units, before this paper settled on the units of §3.3, and have not been re-run; the conclusion is a null either way, which is the one case where the choice of units cannot reverse the answer.

That corpus cannot answer the question anyway. It holds four specimens, so a paired test has an exact two-sided floor of $p = 0.125$ — no result on it can reach significance at any conventional level, a win or a null. One specimen carries 57.7% of all adjudicated not-crack pixels, so the pooled score surface is essentially that specimen's.

Scores computed on the marked area alone can be bought by widening the mask. This is the finding that set the units of this paper, and §3.5 gives it in full: dilating the detector's own output — which reads nothing new from the image — reaches the top of a conventional ranking while pushing the reported crack area fraction to 2.54x the annotator's. A score that a morphological operation can win is not measuring detection.

A control fed the wrong image killed three improvements that looked real. Growing the proposal through a ridge filter, through a local-roughness channel, and by watershed to the crack lip each raised a conventional score substantially. Re-run with the filter computed from a different field, two of the three scored within 0.03 of their real-image version and the third scored below it. All three are in §3.5 with what they cost the area fraction.

Six region-level precision rules validated on 512 px crops were rejected on whole frames. Four of them delete 54–75% of hand-marked crack; the other two change no score. See §2.

Our own corpus cannot referee a detector's false positives. Of the pixels a human actually ruled on across 18 hand-corrected frames, 99.1% are crack — the masks are region assertions over part of each frame, not complete segmentations. A mask that marks every pixel therefore scores 0.995 there on any overlap measure, and no ranking computed on that corpus can be trusted until refereed not-crack area exists. That is the single largest gap in this work and docs/DATASET_RELEASE_SCOPE.md costs out closing it.

6 Measurement practices this forced

Stated as rules because each one followed a specific error of ours.

- Report the score in the units of the answer. Overlap scores discard every pixel

correctly left alone — 98% of the evidence on a 1.80%-crack image — and none of them can state the error a user suffers, which is an area fraction off by some factor. Report labelled %, marked %, and their ratio. That ratio is 1.30x here at the shipped setting, 1.0133x one rung up, and 76.65x on a set this detector fails; no overlap score in any draft of this paper distinguished those three situations as plainly, which is why they are the only units this one uses.

- Never quote plain accuracy without the accuracy of an empty mask. It is 97.42% here and an empty file scores 98.20%. Marking a pixel pays for itself in accuracy only where more than half of what is marked is right, so at low prevalence the two numbers must be printed together or not at all.
- State the whole range the score can move in. A mask with perfect localisation and the best possible width — §4's construction — reaches 98.905% accuracy on these fields against an empty mask's 98.197%. The entire competitive range is 0.708 points. A score quoted without that range cannot be read.
- Report the no-information floor beside every score. Across the sets measured for this project the floor moves by more than a factor of seven, and on our own corpus the refereed pixels are 99.1% crack, so a mask that marks every pixel scores 0.995 there. A bare score is not a result.
- Choose parameters by grouped hold-out, and say what the group is. Regrouping alone moved an AUC by +0.0458 in our data, which is larger than most effects we chased.
- A difference of medians is not a paired difference. One result moved 0.0742 as medians and 0.0000 paired.
- State the resolution of the comparison. Differences of a few thousandths on 26 to 31 images are not results; the mad_k change in §3.2 is reported with a paired test over every image and the count favouring each value, and nothing smaller is claimed anywhere.
- Do not stop a sweep at the value you are validating. Ours ran to the shipped constant and no further, so the claim that the constant was already optimal could not have failed. It did fail when the grid was extended (§3.2), and the original claim was this paper's headline.
- Controls must be able to fail. Every control we built was broken on purpose to check it noticed; two did not and were rewritten. A control fed the wrong image killed three separate improvements in §3.5 that had looked real.
- Check the mask encoding and the bit depth before scoring anything. This set's masks are anti-aliased rather than two-valued and its images are 16-bit; an ordinary 8-bit conversion made the detector find almost nothing and looked like a detector failure rather than a loading bug.
- Score whole fields, not crops of them. Cropping these images around their own cracks moves the reported area fraction from 1.30x to 0.78x with no change to the method, and reverses which operating point is best.
- An oracle must match the arity of the thing it bounds. The constant-width reconstruction in §4 is not an upper bound on an overlap score, and we described it as one until the data said otherwise.

7 What this draft is missing

Listed rather than hidden.

- The evidence is 31 images of one alloy, from a single released collection whose record is one sentence long and names no pixel size, no microscope and no annotation protocol. One crack class out of eighteen targets. That is enough to place the detector and not enough to characterise it, and it is the single largest weakness of this draft.
- A second class from the same collection was measured and removed. The EBC environmental barrier coating (24 fields, 0.26% prevalence) is a ceramic coating rather than a metal and is outside the application. The detector failed on it — it reports 76.65 times the labelled area fraction there — and that failure is not reported above. Removing it also cost Section 4 its strongest form, which compared two sets annotated four times

apart in brush width. Both measurements remain in docs/figures/.

- What was searched for and not found. Public crack segmentation data in this modality is close to non-existent. Ruled out, each for a stated reason: road and pavement sets (CrackForest, DeepCrack, Crack500, CrackSeg9k and the aggregated collections) are the wrong modality; MetalDAM is SEM steel with pixel masks but its five classes are phases and a generic "defect", not cracks; NASA's MicroNet superalloy tasks label precipitates, not cracks; a released Ti-6Al-4V fractography set (SE and BSE, 16 GB) ships images with no masks at all; and the published nickel-superalloy fatigue-crack segmentation work we found is tomography with no data release. A second independent SEM crack set would change this paper more than any further work on the method.
- Pavement results were measured and removed. The detector was scored in overlap units there, reaching F1 0.7932 and IoU
  0.6809 on DeepCrack, ranking first of seven there. It is not reported above because a road-surface benchmark says nothing about micrographs, and the contrast between that ranking and the third-of-seven on SEM is better read as a warning about cross-modality benchmarking than as a result. The numbers remain in docs/figures/public_metrics.json.
- No laboratory results are reported. By choice: the specimen measurements are not being published here. Sections 2 and 5 cite effects measured on that corpus as methodological evidence; a reviewer cannot check those without the data, and they should be read as weaker than the public-dataset numbers.
- The corpus series in Figure 6 carries no stroke width. The instrument that would measure it comparably needs a tile cache not present on this machine.
- The detector's best score in this project is not on cracks. On MetalDAM's voids — compact, dark, high-contrast objects in SEM steel — it reaches accuracy 99.31% and specificity 99.91% there, 1.26 points clear of an empty mask, against 97.42% on
  cracks in the superalloy where an empty mask wins; it reports an area fraction
  0.74x the truth
  against 1.30x. That is what a multi-scale darkness threshold should do, and it bounds how much of this paper is about cracks specifically rather than about dark thin objects. Appendix A gives the full span.
- No timing, and no comparison against a trained network on equal footing. A fair speed claim needs matched hardware, and an earlier attempt found this detector 1.6x slower than a single-pass U-Net at p = 0.0001.
- Novelty is not claimed. Fifty-five framings of this work were checked against prior art and all were already owned. What is offered is measurement.

Data and code availability

The images and masks in Section 3 are the Inconel 718 crack class of a public collection, Zenodo record 23045328, CC BY 4.0 (doi:10.5281/zenodo.23045328). The whole fields are its image/mask directories and the released crops its image_cropped/mask_cropped directories; both are used unmodified except for the 16-bit to 8-bit conversion described in Section 3, which takes the high byte.

Every measurement file this paper draws a number from is listed in docs/INDEX.md with the script that produced it, and the paper itself is built by paper/make_paper.py, which fails rather than renders if a number has no source. The repository is not yet public and no archival DOI or commit hash is quoted here; both are required before submission, and until they exist a reader cannot check any of the above. The hand-corrected micrographs of our own laboratory referred to in Sections 2, 5 and 6 are not in a public release either; docs/DATASET_RELEASE_SCOPE.md measures what releasing them would cost and lists the decisions still open.

References

Implementations of the five classical arms are those in scikit-image [10]; each was run

through this paper's own scoring code, not its authors'.

[1] Otsu, N. (1979). A threshold selection method from gray-level histograms. IEEE Transactions on Systems, Man, and Cybernetics 9(1), 62–66.

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[4] Sato, Y., Nakajima, S., Shiraga, N., Atsumi, H., Yoshida, S., Koller, T., Gerig, G. and Kikinis, R. (1998). Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. Medical Image Analysis 2(2), 143–168.

[5] Meijering, E., Jacob, M., Sarria, J.-C.F., Steiner, P., Hirling, H. and Unser, M. (2004). Design and validation of a tool for neurite tracing and analysis in fluorescence microscopy images. Cytometry Part A 58(2), 167–176.

[6] Olofsson, P., Foody, G.M., Herold, M., Stehman, S.V., Woodcock, C.E. and Wulder, M.A. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment 148, 42–57. doi:10.1016/j.rse.2014.02.015. The closest prior art to Section 3.3. Reporting the error matrix in units of area proportion, and correcting a map-derived area for classifier bias, is standard practice in land-cover accuracy assessment and has been for a decade. What Section 3.3 adds is not the idea but the application to a micrograph area fraction, and a full budget rather than a single bias-adjusted point estimate.

[7] Durmaz, A.R., Müller, M., Lei, B., Thomas, A., Britz, D., Holm, E.A., Eberl, C., Mücklich, F. and Gumbsch, P. (2021). A deep learning approach for complex microstructure inference. Nature Communications 12, 6272. Precedent in this modality for reporting the downstream morphological quantity beside the segmentation score, and for the observation that phase-boundary localisation limits per-object morphology.

[8] ASTM E562-19. *Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count.* ASTM International. The units a materials laboratory reports in, and the accuracy target Section 6 measures against.

[9] Underwood, E.E. and Banerji, K. Quantitative fractography. doi:10.1007/BF02698249. The roughness and profile parameters the measurement side of this tool reports.

[10] van der Walt, S., Schönberger, J.L., Nunez-Iglesias, J., Boulogne, F., Warner, J.D., Yager, N., Gouillart, E. and Yu, T. (2014). scikit-image: image processing in Python. PeerJ 2, e453.

[11] Salvato, D. et al. (2023). Impact of SEM acquisition parameters on the porosity analysis of irradiated U-Mo fuel. Nuclear Materials and Energy. doi:10.1016/j.nme.2023.101494. Published precedent that a segmentation-derived extent metric moves with acquisition settings, which is the effect Section 5 reports for detection mode.

[12] Dahari, A. et al. (2025). Prediction of microstructural representativity from a single image. Advanced Science. doi:10.1002/advs.202414149, arXiv:2410.19568. Bounds the within-image representativity of a phase fraction analytically; it does not estimate between-patch variance over a specimen surface, which is the quantity Section 7 says this corpus cannot supply.

[13] The Inconel 718 crack class used throughout Section 3 is from a public cross-material microstructure collection, Zenodo record 23045328, CC BY 4.0. doi:10.5281/zenodo.23045328.

This list is incomplete. It covers the arms this paper runs and the prior art it has checked itself against. It does not yet contain a survey of learned crack segmentation, and the sentence in Section 3.4 about the published learned state of the art is uncited and

should either be sourced or deleted before submission.

Appendix A Every set measured, as a pixel budget

All five SEM sets this project has scored, in the units the tool reports. Two of them are not crack benchmarks and say so. Percentages are of all image pixels, averaged per image, mad_k chosen leave-one-image-out.

set	lab%	mark%	ok%	bad%	rep/true	acc%	empty%	spec%
Inconel 718, fields	1.80%	2.34%	0.78%	1.56%	1.30x	97.42	98.20	98.41
Inconel 718, crops	4.53%	3.53%	2.11%	1.41%	0.78x	96.17	95.47	98.52
EBC coating, fields	0.26%	20.29%	0.19%	20.10%	76.65x	79.83	99.74	79.85
EBC coating, crops	0.61%	19.70%	0.49%	19.21%	32.54x	80.68	99.39	80.66
MetalDAM voids	1.95%	1.44%	1.35%	0.09%	0.74x	99.31	98.05	99.91

empty% is the accuracy an empty mask scores on that set, which is (100 − lab%). Two rows beat it: MetalDAM by 1.26 points and the Inconel crops by 0.70. Set sizes: Inconel 31 fields and 26 crops, EBC 24 fields and 8 crops, MetalDAM 16.

The span is the point. The same detector, unchanged, reports an area fraction that is 0.74 times the truth on compact voids in steel and 76.65 times the truth on a coating at 0.26% prevalence. On the alloy it is 1.30x. Anyone quoting a single accuracy figure for this detector is quoting one point on that range.

MetalDAM is the most revealing row and it is not a crack result. Its targets are voids: compact, dark, high-contrast. The detector reaches accuracy 99.31% and specificity 99.91% there — the highest figures anywhere in this project, and 1.26 points clear of an empty mask, the widest such margin we have — against 97.42% on cracks in the superalloy, where an empty mask wins. Its best performance is on the objects that are easiest to be dark, which is consistent with what it is: a multi-scale darkness threshold. Cracks are hard for it because they are thin and low in contrast, not because they are cracks.

The two pavement sets are absent. They were measured on overlap scores before being removed from this paper as the wrong modality, and their downloads were not retained, so no pixel budget was ever computed for them.

Appendix B. The detector, end to end



Figure 7. The shipped pipeline. Five constants are read from analysis/detector_overrides.py when the plate is drawn; mad_k and the two size floors are typed into the generator and are not yet under that check. Pass 2 and the hand-correction layer are stages of the application and are not in the scored path.